

Mod-2 period-6 and mod-3 anti-period laws for the recurrence family $(6, 9, 0, -1)$, with a matrix-functional generalization (Lean 4 machine-checked claim of record)

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Abstract

We record two theorems about the order-4 integer recurrence $c(n+4) = 6c(n+3) + 9c(n+2) - c(n)$ — the recurrence of the matching counts of the $3 \times n$ cylindrical grid (OEIS entry A033515). Theorem 1: every integer sequence satisfying this recurrence has period dividing 6 modulo 2, and satisfies the anti-period relation $c(n+4) \equiv -c(n) \pmod{3}$. Theorem 2: every matrix-power functional of an 8×8 integer matrix annihilated by $x^4 - 6x^3 - 9x^2 + 1$ inherits the same recurrence, uniformly in the vector and the coordinate. All proofs are machine-checked by the Lean 4 kernel (0 sorry; axioms within {propext, Quot.sound, Classical.choice}). A structured novelty search found no occupying record. The artifact package is deposited on Zenodo. This note is a priority claim of record, not a full paper.

1 Introduction

The graph $C_3 \square P_n$ — the $3 \times n$ grid wrapped around a cylinder, also called the stacked prism graph — has a classical matching count, OEIS entry A033515 [8], whose record lists the order-4 recurrence $c(n+4) = 6c(n+3) + 9c(n+2) - c(n)$. Numerical probes of three fixed three-hole defect configurations of the same grid found (computationally; see Section 6) counting sequences that satisfy the same recurrence with different initial values.

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This note records two theorems about that recurrence family, produced by AI4Math, a highly automated LLM-plus-kernel pipeline in which the Lean 4 kernel is the sole arbiter of correctness. The present text is a priority claim of record — a short deposit note accompanying a Zenodo artifact package; a fuller narrative paper is prepared separately through the pipeline’s paper lane. Every theorem stated below is a *complete proof* in the pipeline’s grading: it compiles with 0 **sorry**, its axioms stay within the standard whitelist, and the statement was frozen before proving.

2 Statement

The formal development quantifies over an *arbitrary* integer sequence satisfying the recurrence, so Theorem 1 covers the whole family (all initial values) at once; congruences use mathlib’s `Int.ModEq` notation `[ZMOD m]`. The listings below reproduce the frozen theorem signatures verbatim (byte-identical excerpts of the frozen statement files `04-statement-p1.lean` and `04-statement-p2.lean`; the files’ placeholder **sorry** bodies are not reproduced; the Chinese docstrings omitted here remain in the package’s `proofs/` directory).

Theorem 1 (recurrence-family congruence law). *Let $c : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy $c(n+4) = 6c(n+3) + 9c(n+2) - c(n)$ for all n . Then, for all n ,*

$$c(n+6) \equiv c(n) \pmod{2} \quad \text{and} \quad c(n+4) \equiv -c(n) \pmod{3}.$$

Listing 1: Formal statement of Theorem 1 (verbatim excerpt, lines 11–13 of the frozen `formalized/04-statement-p1.lean`, sha256 4740b9d5...e733bc).

```

1 theorem cyl3_master (c : ℕ → ℤ)
2   (h : ∀ n, c (n+4) = 6 * c (n+3) + 9 * c (n+2) - c n) :
3   (∀ n, c (n+6) ≡ c n [ZMOD 2]) ∧ (∀ n, c (n+4) ≡ -c n [ZMOD
    3]) := by

```

Theorem 2 (matrix-level inheritance). *Let T be an 8×8 integer matrix with $T^4 = 6T^3 + 9T^2 - I$, let v be an integer vector of length 8, and let i be a coordinate. Then the sequence $a(n) = ((T^n)v)_i$ satisfies the same recurrence: for all n ,*

$$a(n+4) = 6a(n+3) + 9a(n+2) - a(n).$$

Listing 2: Formal statement of Theorem 2 (verbatim excerpt, lines 13–18 of the frozen `formalized/04-statement-p2.lean`, sha256 8cac6fa4...c1d7fe).

```

1 theorem cyl3_matrix_family {T : Matrix (Fin 8) (Fin 8) ℤ}
2   (v : Fin 8 → ℤ) (i : Fin 8)
3   (hT : T ^ 4 = 6 • T ^ 3 + 9 • T ^ 2 - (1 : Matrix (Fin 8) (Fin
4     8) ℤ))
5   (n : ℕ) :
6   (T ^ (n+4)).mulVec v i
   = 6 * (T ^ (n+3)).mulVec v i + 9 * (T ^ (n+2)).mulVec v i -
   (T ^ n).mulVec v i := by

```

3 Proof

The complete machine proofs are reproduced below verbatim: Listing 3 is the entire frozen final development `final/04-proof-complete.lean` (69 lines, sha256 566e9e1d...c5ae8), mechanically spliced byte-for-byte. Its Chinese header comments are pipeline provenance notes (route sketches and source pointers), not mathematical content. A narrative prose proof is not the duty of this note; it belongs to the full paper lane.

Listing 3: Complete development, verbatim: lines 1–69 (the whole file) of the frozen `final/04-proof-complete.lean`, sha256 566e9e1d...c5ae8.

```

1 import Mathlib.Data.Int.ModEq
2 import Mathlib.Data.Matrix.Mul
3
4 /-!
5 # 04-proof-complete (军政部 r1)
6
7 两定理定理头与 tasks/20260926-cyl3-pipeline/formalized/04-statement
8   -p{1,2}.lean
9 冻结文本逐字一致；本文件只写 proof body。AI 生成 (dept-04-prove,
10 node=anthropic/a6api-main/kimi-k3), kernel 实编裁决为准。
11
12 - P1 路线 (03 同型 r2 实证形态) : Int.modEq_iff_dvd 显式 dvd 证人 +
13   ring;
14   模 2 由 `c (n+4) ≡ c (n+2) + c n` 在 n 与 n+2 两处展开后
15   add_right/trans 收束;
16   模 3 反周期一步证人 (6、9 被 3 整除)。
17 - P2 路线 (EdgemidSmoke03r3.lean edgemid_matrix_family 同型临摹, 系
18   数 -6,9,1) :

```

```

15   hgen = pow_add + hT + Matrix.mul_sub/mul_add + mul_smul_comm×2
      + Matrix.mul_one
16   + ← pow_add×2; mulVec 线性 = Matrix.sub_mulVec/add_mulVec +
      smul_mulVec×2;
17   收尾 first | rfl | simp only [smul_eq_mul] | ... (03 r3 已编译验
      证)。
18 -/
19
20 /-- P1 (`cyl3_master`, 修正档=稳妥): 任一满足四阶线性递推
21 `c (n+4) = 6 * c (n+3) + 9 * c (n+2) - c n` 的整数序列,
22 模 2 以 6 为周期, 且模 3 有反周期 `c (n+4) ≡ -c n [ZMOD 3]`
23 (推论: `3 | c n ⇔ 3 | c (n % 4)` 型整除刻画与基线奇偶模式)。
24 人话语义: 带缺陷圆柱格  $C_3 \times P_n$  ( $3 \times n$ , 三种固定孔位) 匹配计数族的
25 统一同余律——无孔基线 (OEIS A033515) 与同行相邻/对角/三角三种
26 三孔缺陷序列同满足递推  $-(6, 9, 0, 1)$ , 本定理对该递推整族一次覆盖。
27 题源: tasks/20260926-beian-select/candidates/04-cylinder-3holes-
      review.md (收窄命题 P1)。 -/
28 theorem cyl3_master (c : ℕ → ℤ)
29   (h : ∀ n, c (n+4) = 6 * c (n+3) + 9 * c (n+2) - c n) :
30   (∀ n, c (n+6) ≡ c n [ZMOD 2]) ∧ (∀ n, c (n+4) ≡ -c n [ZMOD
31   3]) := by
32   have key : ∀ m, c (m+4) ≡ c (m+2) + c m [ZMOD 2] := by
33     intro m
34     refine Int.modEq_iff_dvd.2 ⟨-(3 * c (m+3) + 4 * c (m+2) - c m)
35     , ?_⟩
36     rw [h m]
37     ring
38     refine ⟨fun n => ?_, fun n => ?_⟩
39     · have h1 : c (n+4) ≡ c (n+2) + c n [ZMOD 2] := key n
40       have h2 : c (n+6) ≡ c (n+4) + c (n+2) [ZMOD 2] := key (n+2)
41       have h3 : c (n+6) ≡ (c (n+2) + c n) + c (n+2) [ZMOD 2] := h2.
42         trans (h1.add_right _)
43       exact h3.trans (Int.modEq_iff_dvd.2 ⟨-c (n+2), by ring⟩)
44     · refine Int.modEq_iff_dvd.2 ⟨-(2 * c (n+3) + 3 * c (n+2)), ?_⟩
45       rw [h n]
46       ring
47
48 /-- P2 (`cyl3_matrix_family`, 挑战档):  $8 \times 8$  整数转移矩阵 `T` 一旦被
49 四次多项式零化 ( $T^4 = 6 \bullet T^3 + 9 \bullet T^2 - 1$ ), 其任意矩阵幂
50 泛函 `(T^(n+k)).mulVec v i` 都满足同一个四阶递推
51 `a (n+4) = 6 * a (n+3) + 9 * a (n+2) - a n`——一条引理统一三种
52 固定孔位缺陷计数在转移矩阵层的共有结构 (矩阵泛函层统一递推定理)。
53 人话语义: 这是转移矩阵法 (Lundow 1996/1998 口径) 的泛函形式化;
54 「该矩阵确实等于缺陷匹配计数」的组合语义桥不进 kernel,
55 论文/报告层按主张降级如实声明, 不得拔高为计数本体定理。

```

```

53 题源: tasks/20260926-beian-select/candidates/04-cylinder-3holes-
    review.md (收窄命题 P2)。 -/
54 theorem cyl3_matrix_family {T : Matrix (Fin 8) (Fin 8) ℤ}
55   (v : Fin 8 → ℤ) (i : Fin 8)
56   (hT : T ^ 4 = 6 • T ^ 3 + 9 • T ^ 2 - (1 : Matrix (Fin 8) (Fin
    8) ℤ))
57   (n : ℕ) :
58   (T ^ (n+4)).mulVec v i
59   = 6 * (T ^ (n+3)).mulVec v i + 9 * (T ^ (n+2)).mulVec v i -
    (T ^ n).mulVec v i := by
60 have hgen : T ^ (n+4)
61   = 6 • T ^ (n+3) + 9 • T ^ (n+2) - (T^n : Matrix (Fin 8) (Fin
    8) ℤ) := by
62   rw [pow_add, hT, Matrix.mul_sub, Matrix.mul_add,
63     mul_smul_comm, mul_smul_comm,
64     Matrix.mul_one, ← pow_add, ← pow_add]
65 -- 收尾: rw 自带 rfl 不足以穿透 Pi 逐点作用 (r2 实测 unsolved
    goals),
66 -- 显式 `rfl` tactic 以全 defeq 收官 (r1 级联实测首支即中, 故仅留
    此一支)。
67 rw [hgen, Matrix.sub_mulVec, Matrix.add_mulVec,
68   Matrix.smul_mulVec, Matrix.smul_mulVec]
69 rfl

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (leanprover/lean4:v4.34.0) [1], mathlib4 v4.34.0 (rev 5ed29652) [2].
- Statement integrity: frozen statements formalized/04-statement-p1.lean, sha256 4740b9d53e97360c49d0eefe97a7abc470134acf8e61273963d7a55de8e733bc, and formalized/04-statement-p2.lean, sha256 8cac6fa4ec068dc913ff90769bda34333bddd9c (LF-normalized); final development final/04-proof-complete.lean, sha256 566e9e1d7053602eda7853943f582e3d3726009c32f9dc4b6ce3ca8ee19c5ae8; symmetric difference of the two theorem heads (up to := by) against the frozen basis: zero (verbatim).
- Kernel check: strict re-compilation (scripts/lean-verify, strict mode) exit 0 with empty diagnostics; a byte-level scan of the whole file (comments included) finds 0 occurrences of sorry/admit. Axiom audit (sidecar re-compilation), verbatim output (audit/audit-axioms-sidecar.log):

```
'cyl3_master' depends on axioms: [propext, Classical.choice, Quot.sound]
'cyl3_matrix_family' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Both are within the standard whitelist {propext, Quot.sound, Classical.choice}; no `sorryAx`.

- Degeneracy probing: the well-definedness gate ran 14 bare-tactic probes (7 against each statement: `nlinarith`, `norm_num`, `simp`, `ring_nf`, `positivity`, `omega`, `linarith`); all failed as required, so neither theorem is closable by a bare decision procedure.
- Audit chain: gate-2 well-definedness compilation plus degeneracy probing, statement freeze with sha256 pins, gate-3 final strict compilation and axiom audit (`audit/final-04-audit.md`: all four checks pass; grading *complete proof* for both theorems). Evidence originals are in the artifact package `audit/`.

5 Novelty evidence

The pipeline’s C9 novelty re-review (2026-09-26; independent re-query before deposit) adjudicated, per proposition: (A) the three fixed three-hole defect sequences — *no occupying record found*, established and strengthened (the sequences were recomputed by an independent enumerator implementation, then probed on OEIS through 19 sequence-shape queries and 6 keyword groups and on the literature channels below); (P1) Theorem 1 — *no occupying record found*, positioned as a congruence-law observation accompanying the new sequences (new-sequence tier; an elementary Pisano-type instance of classical modular-period theory, not new mathematics); (P2) Theorem 2 — *no occupying record found*, positioned as a library-lemma-level first formalization; (P3) the full combinatorial semantics — occupied at the literature layer [6, 7, 8, 9], hence not a contribution of this note. This is search evidence, not a claim to new mathematics. Query-channel summary (the full query log, including zero-hit evidence files, is copied verbatim into `audit/c9-record.md` of this package):

channel		query summary	result
OEIS layer	sequence	26 queries (19 sequence-shape probes of the defect families plus recurrence-text and generating-function probes) + 2 record reads; calibration query hits A033515 (channel works)	defect variants: all null
arXiv		8 query groups (incl. 3 http trial-and-error); nearest lines arXiv:2109.12716 [3], cond-mat/0611449 [4], 2406.05750 [5] read and delineated	no occupant
OpenAlex		5 broad + 3 title-exact queries	no occupant
zbMATH		4 query groups	no occupant; 3 records license-masked (gap)
Semantic Scholar		1 query	HTTP 429, not retried (gap)
general (Bing/DDG)	web	5 fetches	channel degraded (gap)
CNKI		not searched (login required)	gap
formal libraries		mathlib / compfiles / sequencelib probes (Hosoya, monomer, A033515, ...)	zero hits; A033515 absent from sequencelib

Recorded residual gaps, verbatim in substance from the re-review: CNKI not searched (login required); one Semantic Scholar query rate-limited (HTTP 429) and not retried; three zbMATH records license-masked and unreadable; the general web search layer degraded (Bing localization pollution; DuckDuckGo blocked by DNS pollution and proxy failure).

6 Scope and limitations

Grading boundary, per the final audit’s transfer wording: the two theorems are *complete proofs*, strictly as statements about the recurrence family (an arbitrary integer sequence c) and about matrix-power functionals. The kernel did *not* prove that any concrete defect matching-count sequence equals this recurrence or this matrix: the statement “the three fixed three-hole de-

fect counting sequences satisfy the recurrence $(6, 9, 0, -1)$ ” rests on the OEIS record and the transfer-matrix literature (A033515 and A287428 are registered, not new [8, 9, 6, 7]) plus a local computational probe (range $n \leq 14$, two independently implemented enumerators agreeing term by term); it is reported at the computational-conjecture layer and is not part of the kernel-proved content. Probe values, copied verbatim from the source records:

family (fixed three-hole configurations)	first values
baseline, no holes (OEIS A033515, registered)	1, 4, 32, 228, 1655, 11978, 86731, 627960
row3: three adjacent holes in one row	22, 135, 1006, 7251, 52538, 380352, 2753948, 19939605
diag3: three diagonal holes	13, 83, 615, 4437, 32144, 232714, 1684965, 12199779
tri3: three triangular holes	3, 19, 141, 1017, 7368, 53342, 386223, 2796399

Further boundaries. Theorem 2 is an elementary consequence of the annihilation hypothesis (Cayley–Hamilton-type reasoning); the contribution is the packaging as one reusable kernel-checked lemma — a library-lemma-level first formalization, not new mathematics. A fourth configuration with a moving hole was recorded with a sixth-order recurrence fit; it stays at record level and is not a contribution claim. The novelty evidence above is negative search evidence with the stated residual gaps; it supports no first-discovery claim beyond the stated tier. The source pipeline report is marked “pending gate-4 human sign-off”; this deposit is issued under the depositing author’s explicit start instruction of 2026-09-27 (recorded in this package’s card). This note contains no literature survey; it is a deposit record, not a survey article.

Data availability

Artifact package: Zenodo deposit (DOI to be reserved at upload; see UPLOAD.md in this package), with a public mirror on GitHub at <https://github.com/Aurora0134/ai4math-claims>. Note under CC BY 4.0, code under MIT.

Use of generative AI

(a) Stage mapping: candidate-proposition generation, formalization into Lean statements, proof search, and text drafting (including this note) were performed by AI (LLM) stages of the AI4Math pipeline; topic adjudication, statement-semantics verification and gate sign-off were human. (b)

Model/node/budget (numbers copied verbatim from the source task ledger `tasks/20260926-cyl3-pipeline/budget.log`): a single resolved model node, wire-id `anthropic/a6api-main/kimi-k3` (resolution level L2 relay), no cross-node switching; the source task totals llm-call 1/6 (the roundtrip semantic judge), kernel compilations 12/24, diagnostic probes 0/4; the separate full-paper lane used LaTeX compilations 6/10; the C9 exclusivity re-review used 0 LLM calls and 60 retrieval requests. This claim-note stage used llm-call 0 of its budget 4, and 5 LaTeX compilations of its budget 6 (counting the final rebuild that embedded this number; ledger `claims/cyl3-mod23/budget.log`). (c) Final arbiter: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only `{propxt, Quot.sound, Classical.choice}` for both theorems. (d) The human author reviewed and is responsible for all content; the AI system is not an author.

Author contributions

CRediT-style: the human author was responsible for topic selection and adjudication, verification of statement semantics, gate sign-off, and the decision to deposit; the AI4Math pipeline (an LLM system) performed proposition generation, proof search, and text drafting. The AI system is not listed as an author.

References

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